

SD-WAN Resilience Test Report

Automatic failover of site BR8 when data center DC1 loses its internet access

Prepared for: [Customer name] · Test date: September 24, 2026 · Environment: Prisma SD-WAN lab (sites BR8, DC1, DC2)

1. Executive summary

Using real, repeated outages, we verified how site BR8 reacts when the internet access of data center DC1 becomes unavailable. Across all runs, **failover to the backup data center DC2 was triggered automatically every time**, with no manual intervention, and traffic returned to DC1 once the link was restored. The resilience of the architecture is therefore demonstrated. The measurements also quantify the exact impact experienced by users and highlight clear optimization opportunities.

100% of outages followed by automatic failover	< 2 s to decide to fail over to DC2	≈ 8.6 s average outage (7.7 to 9.3 s)	~3 s for TCP applications to reconnect
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Key takeaway. Failure detection is fast (under 2 seconds), but traffic takes about 8 seconds to be fully restored, mainly because of the return path from the data center. This is the delay users actually feel: an ongoing voice call is interrupted, and application sessions to DC1 are dropped, then re-established automatically. Failing back to DC1, on the other hand, is transparent for established sessions.

2. Context and methodology

2.1 Tested architecture

Site BR8 has two internet circuits (INET1 and INET2) and encrypted SD-WAN tunnels to two data centers. DC1 is the preferred data center, DC2 the backup. BR8 has no MPLS link: when DC1's internet access goes down, the only alternative path goes through DC2. During testing, the path policy was refined to reference only the paths actually available to BR8.

2.2 Test procedure

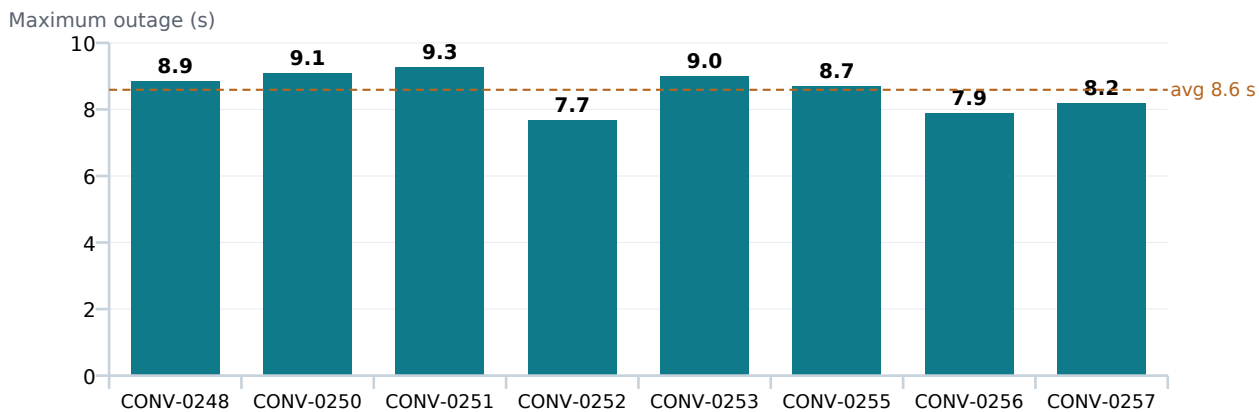
Every test follows the same protocol: a continuous measurement flow is sent from BR8 to a server in DC1 (50 packets per second, echoed back, which makes it possible to measure loss separately in each direction). DC1's internet access is then shut down on the lab router for about one minute, and restored. Depending on the run, a +200 ms latency degradation was applied beforehand, and the link was restored either during the measurement or after it was stopped. In parallel, simulated voice calls (G.711, 30 s) and TCP applications ran continuously to measure the real application impact.

Tests were generated and measured with the Stigix platform, orchestrated and analyzed with the Claude assistant connected to the tools through MCP. Path decisions were cross-checked with the Prisma SD-WAN Flow Browser. All timestamps in this report are UTC.

3. Results

3.1 Outage duration during failover

Eight failovers were measured over the course of the day. The maximum outage observed by the measurement flow is remarkably stable, between 7.7 and 9.3 seconds, with an average of 8.6 seconds. This consistency points to a deterministic detection and convergence mechanism rather than random behavior.



Maximum outage measured for each failover test (seconds).

3.2 Outage breakdown

In the last two runs, the exact timestamp of the link shutdown made it possible to separate the failover decision from the actual restoration of traffic:

Measurement	Test CONV-0256	Test CONV-0257
Link shutdown (exact timestamp)	20:05:03.599	20:11:10.148
First path decision	+1.4 s	+0.9 s
Failover to DC2	+1.8 s	+1.6 s
Maximum outage	7.9 s	8.2 s
Loss BR8 → DC1 (upstream)	2.9 s	3.0 s
Loss DC1 → BR8 (downstream)	5.0 s	5.3 s

The failover decision is made in under 2 seconds, but traffic only flows normally about 8 seconds after the shutdown. Most importantly, **the return direction (DC1 to BR8) recovers about 2 seconds later than the forward direction**: BR8 fails over quickly, but replies from the data center take longer to find the path through DC2. Data-center-side convergence is therefore the main lever for improvement.

3.3 Latency degradation without outage

A 200 ms latency increase on DC1's internet access, sustained for 30 seconds, triggered **no failover** in any of the five runs concerned. Traffic stayed on the degraded path. This behavior is consistent with an availability-based policy; it should be confirmed whether it matches business expectations.

3.4 Application impact

Traffic type	During the outage	On failback to DC1
UDP measurement flow	Outage of 7.7 to 9.3 s	Transparent
Voice to DC1	Call in progress at the moment of the outage: 26% loss, MOS 1.48 (unusable). Call started during the outage via DC2: perfect (MOS 4.4)	No impact (MOS 4.4)
TCP applications to DC1	Sessions dropped, then automatically re-established in ~3 s via DC2	Sessions preserved, single latency spike (759 ms)
TCP to other sites (control)	No impact	No impact
Voice to other branches	Micro-interruption of about 0.5 s (MOS 4.21 to 4.31)	No impact

The real user impact does not come from the path change itself but from **application reconnection**: ongoing sessions to DC1 do not survive the 8-second outage, whereas failback to DC1 happens without any interruption. Voice impact depends entirely on timing: only a call in progress at that exact moment is affected.

3.5 Failback to the primary data center

After the link is restored, new flows are steered back to DC1 within about ten seconds. In the last run, failback was timed precisely: link restored at 20:13:40.7, first flows sent to DC1 at 20:13:50.9, but first replies received only at 20:13:57.5. **For about 6.6 seconds, new flows were therefore sent to DC1 before that path was actually carrying traffic.**

In addition, with the latest software version installed on BR8, flows already established on DC2 stayed there until they ended, while new flows returned to DC1. This behavior avoids a second disruption of ongoing sessions; it should be confirmed and documented as expected behavior.

4. Recommendations

Priority	Recommendation	Expected benefit
High	Analyze return-path convergence on the data center side (DC1/DC2 routing toward BR8), which accounts for most of the gap between decision (< 2 s) and restoration (~8 s).	Shorter outage perceived by users
High	Review failback timers to avoid sending flows to a path that is not yet operational (~6.6 s observed).	Removal of the micro-outage on failback
Medium	Decide whether performance degradation (latency, loss) should trigger failover, and adjust the policy accordingly (quality-of-service criteria).	Alignment with business expectations
Medium	For voice and critical applications, evaluate the available real-time protection features and verify the automatic reconnection logic of applications.	Better perceived continuity of calls and sessions
Ongoing	Include these failover tests in the change process: replay them after every software upgrade or policy change, and compare against the baseline values in this report.	Measurable non-regression

Baseline values. Failover decision < 2 s · average outage 8.6 s (7.7–9.3 s) · TCP reconnection ~3 s · failback to DC1 transparent for established sessions. These values can serve as an internal service commitment and as a comparison baseline for future tests.

Appendix A. Failover test details

Loss percentages depend on each test's duration (the same outage weighs more in a short test); the maximum outage is the most comparable indicator from one test to another.

Test	Start (UTC)	Scenario	Max. outage	Upstream / downstream loss
CONV-0248	11:01	+200 ms then shutdown, failback during test	8.9 s	1.2% / 2.0%
CONV-0250	12:39	+200 ms then shutdown, failback during test	9.1 s	2.0% / 3.3%
CONV-0251	16:27	Shutdown, stopped during outage (manual test)	9.3 s	2.4% / 2.3%
CONV-0252	16:36	+200 ms then shutdown, failback during test	7.7 s	1.7% / 2.4%
CONV-0253	17:08	Same, with voice and TCP monitoring	9.0 s	2.0% / 3.1%
CONV-0255	17:28	Shutdown, stopped during outage	8.7 s	2.7% / 7.4%
CONV-0256	20:04	Shutdown, failback during test (new version)	7.9 s	1.4% / 2.4%
CONV-0257	20:10	Shutdown, stopped during outage (new version)	8.2 s	3.6% / 6.7%

Appendix B. Detailed failover timeline (CONV-0256)

Time (UTC)	Event
20:04:23.9	Measurement flow starts on BR8-INET2 → DC1-INET
20:05:03.599	DC1 internet access shut down
20:05:04.994	First new decision: BR8-INET1 → DC1-INET attempted (DC1 still unreachable)
20:05:05.283	Another tunnel to DC1-INET attempted
20:05:05.433	Failover to DC2 (DC2-INET ↔ BR8-INET1), 1.8 s after the shutdown
~20:05:11	Traffic fully restored (maximum outage measured: 7.9 s)
20:06:27.954	DC1 internet access restored
20:07:34	End of test; the established flow stayed on DC2 until it ended

Note: times marked "~" are inferred from measurements (maximum outage), not from a direct timestamp.